Step 6 — Signal & Connection Engine

Goal

Simulate logical broadcast signal flow, not actual video.

Think of a signal as a packet of metadata describing a video stream.

Example:

Camera

↓

Video Signal

↓

Router

↓

Encoder

↓

Viewer

Core Concept

A connection carries a Signal, not pixels.

Example Signal

Resolution : 1920x1080

FPS : 60

Codec : RAW

Bitrate : 3 Gbps

Status : ACTIVE

The Router or Encoder modifies this signal.

Connection Model

Every connection links:

Source Device

↓

Source Port

↓

Target Port

↓

Target Device

Instead of:

Camera → Encoder

We'll have:

Connection

sourceDevice

sourcePort

targetDevice

targetPort

status

This is much more flexible.

Ports

Every device owns ports.

Example

Camera

OUT1

Router

IN1

IN2

IN3

OUT1

OUT2

OUT3

Encoder

IN1

OUT1

Signal Object ⭐

Instead of passing strings, create a Signal object.

It contains:

Signal

signalId

resolution

fps

bitrate

codec

quality

latency

status

This object flows through the network.

Signal Lifecycle

Camera

creates Signal

↓

Router

routes Signal

↓

Encoder

encodes Signal

↓

Viewer

displays Signal

Each device may modify the signal.

Signal States

ACTIVE

DEGRADED

LOST

BLOCKED

Example:

Camera disconnects

↓

Signal becomes

LOST

Router forwards

LOST

Viewer displays

No Signal

Signal Quality

Every signal has a quality score.

100%

↓

95%

↓

80%

↓

60%

↓

0%

This allows gradual degradation instead of immediate failure.

Propagation

Signals always move downstream.

Example

Camera

↓

Router

↓

Encoder

↓

Viewer

Every simulation tick

Camera

↓

Generate Signal

↓

Router

↓

Forward Signal

↓

Encoder

↓

Modify Signal

↓

Viewer

↓

Consume Signal

Router Switching ⭐

This is one of the coolest features.

Example

Initially

Camera 1

↓

Router IN1

↓

OUT1

↓

Encoder

User changes routing

Camera 2

↓

Router IN2

↓

OUT1

↓

Encoder

Only the Router changes.

Everything downstream automatically receives Camera 2.

Multiple Inputs

Later

Camera 1

Camera 2

Replay

Graphics

↓

Router

The Router decides which input goes to which output.

Signal Processing

Different devices modify signals differently.

Camera

Creates signal.

Router

Routes signal.

Encoder

Changes

RAW

↓

H264

↓

Bitrate Reduced

Decoder

Changes

H264

↓

RAW

Viewer

Consumes signal.

Failure Propagation ⭐

Example

Camera power failure.

Camera

↓

Signal LOST

↓

Router

↓

Output LOST

↓

Encoder

↓

Black Frame

↓

Viewer

↓

No Video

↓

Alarm

Failures naturally propagate through the topology.

Connection Validation

Before simulation starts

Check

- Connected?
- Input already used?
- Output available?
- Circular connection?

Reject invalid topologies.

Connection States

CONNECTED

DISCONNECTED

DISABLED

BROKEN

Dynamic Re-routing

Future feature.

If

Camera 1

↓

FAILED

Automatically

Camera 2

↓

Take Over

Great extension for V2.

Connection Registry

Instead of searching

Maintain

Connection Registry

Camera → Router

Router → Encoder

Encoder → Viewer

The Signal Engine traverses this graph every tick.

Signal Flow

Every simulation tick

Generate Signals

↓

Move Through Connections

↓

Modify Signals

↓

Consume Signals

↓

Generate Events

↓

Update Dashboard

Final Architecture

Signal Engine

│

├── Signal Generator

├── Connection Manager

├── Port Manager

├── Signal Processor

├── Signal Validator

└── Propagation Manager

One Addition (Very Important)

Device Capabilities

Instead of hardcoding behavior, every device declares its capabilities.

Example:

Camera

Produces Signal ✓

Consumes Signal ✗

Modifies Signal ✗

Routes Signal ✗

Router

Produces Signal ✗

Consumes Signal ✓

Modifies Signal ✗

Routes Signal ✓

Encoder

Produces Signal ✗

Consumes Signal ✓

Modifies Signal ✓

Routes Signal ✗

Viewer

Produces Signal ✗

Consumes Signal ✓

Displays Signal ✓

This makes the engine generic.

The Simulation Engine doesn't need to know what a device is—it only asks what capabilities it has.